RAJASTHAN ROYALS · SupeRR SELECTOR SEASON 2

Identifying the Most Reliable Players Under Pressure

Round 1 Methodology Report

A reproducible, delivery-level framework that measures how batters and bowlers perform in high-pressure situations relative to their own baseline, across IPL and T20I cricket (2022–2026).

Eligible pool: 112 batters, 171 bowlers · 867 matches · 200,961 deliveries analysed · IPL + T20I combined, format-adjusted baselines

Confidential — prepared on supplied proprietary competition data. Not for distribution.

1 · Executive Summary

We define **pressure** as a continuous, per-delivery intensity (0–1) built from the match situation *before* each ball is bowled — innings phase, wickets already lost, balls remaining and, in run-chases, the required run rate and how alive the contest is. Crucially we build **two separate, asymmetric models**: a batting team is under pressure in hard chases and at the death, whereas a bowling team is under pressure when batters are set and scoring freely. Every player is judged against a **context-matched baseline** (their own non-pressure level in the same innings phases) and scored on an **absolute-leaning blend** — 50% absolute pressure performance, 30% uplift versus baseline, 20% match-to-match consistency — with empirical-Bayes shrinkage so small samples cannot spike. Rankings are stress-tested across nine model configurations; the **reliability** column is the share in which a player stays in the top 10. These lists are the **primary statistical result and apply no nationality filter**; an optional IPL-availability view is given in §15.

Top 10 Batters Under Pressure

#	Batter	Score	Rel.	Pr. balls	Pr. mts	Pressure SR	Base SR	SR upl.	Pr. avg
1	**Matthew Wade**	88.6	67%	229	19	162	110	+52	53
2	**Naman Dhir**	87.2	89%	267	25	193	152	+41	32
3	**Babar Azam**	82.6	67%	114	8	193	136	+57	44
4	**Rohit Sharma**	82.5	33%	82	5	194	142	+52	80
5	**Rishabh Pant**	82.1	89%	339	31	182	137	+46	33
6	**Mohammad Rizwan**	82.1	44%	117	9	174	126	+47	41
7	**Shai Hope**	79.3	56%	116	12	178	139	+40	34
8	**Harry Tector**	79.2	56%	182	16	162	114	+48	30
9	**Fakhar Zaman**	78.8	67%	129	12	171	124	+48	32
10	**Shubman Gill**	78.8	33%	202	17	193	155	+38	39

Top 10 Bowlers Under Pressure

#	Bowler	Type	Score	Rel.	Pr. balls	Pr. mts	Pr. wkts	Pressure econ	Base econ	Wkts/ over
1	**Haris Rauf**	Pace	95.3	100%	283	29	28	8.31	9.16	0.59
2	**Craig Young**	Pace	89.9	78%	98	11	9	8.45	8.34	0.55
3	**Ravi Bishnoi**	Spin	88.6	78%	223	28	18	9.09	8.86	0.48
4	**Wanindu Hasaranga**	Spin	88.3	78%	432	37	33	7.79	7.96	0.46
5	**Mitchell Starc**	Pace	86.6	89%	171	20	19	10.11	9.68	0.67
6	**Moeen Ali**	Spin	86.1	56%	73	11	8	6.66	8.73	0.66
7	**Vaibhav Arora**	Pace	85.9	56%	171	22	15	10.14	10.34	0.53
8	**Digvesh Rathi**	Spin	85.1	56%	146	14	9	8.26	9.65	0.37
9	**Mukesh Choudhary**	Pace	84.5	67%	97	13	7	9.40	10.94	0.43
10	**Nathan Ellis**	Pace	82.4	89%	193	21	13	8.83	8.72	0.40

Rel. = reliability (of 9 configs). Pr. = pressure. SR upl. = pressure SR minus own baseline SR. Wkts credited to the bowler only (run-outs excluded).

2 · Problem Statement & Approach

Round 1 asks us to build a data-driven system that measures how batters and bowlers perform in high-pressure situations compared with their usual level, and to rank the players who deliver most consistently. The definition of pressure is left to us. Our guiding principles were:

- **Reliability over reputation.** Every pick is derived from the supplied ball-by-ball data; no player list is hard-coded.
- **Self-referenced.** A high pressure strike rate is only credited relative to that batter's own baseline — we reward players who *maintain or raise* their level when it matters.
- **No future leakage.** Pressure at a delivery uses only information available up to that ball; the match result is never used to label earlier deliveries.
- **Honest about the data.** Where a quantity (e.g. exact team totals) cannot be reconstructed reliably, we say so and avoid building on it.
- **Robustness first.** A single formula is not trusted; the answer must survive reasonable changes to thresholds, weights and the pressure definition itself.

3 · Dataset

Seven supplied files. The core is **ball_by_ball.csv** (200,961 main-innings deliveries after excluding 6 super-over innings), joined to match metadata (883 matches, 867 with ball data), competitions (141; 5 IPL seasons 2022–2026), teams, playing XIs and competition–team–player mappings. Coverage splits roughly evenly: ~397 T20 matches (371 IPL) and ~470 T20I. All joins verified; the delivery key (match, innings, ball) is unique.

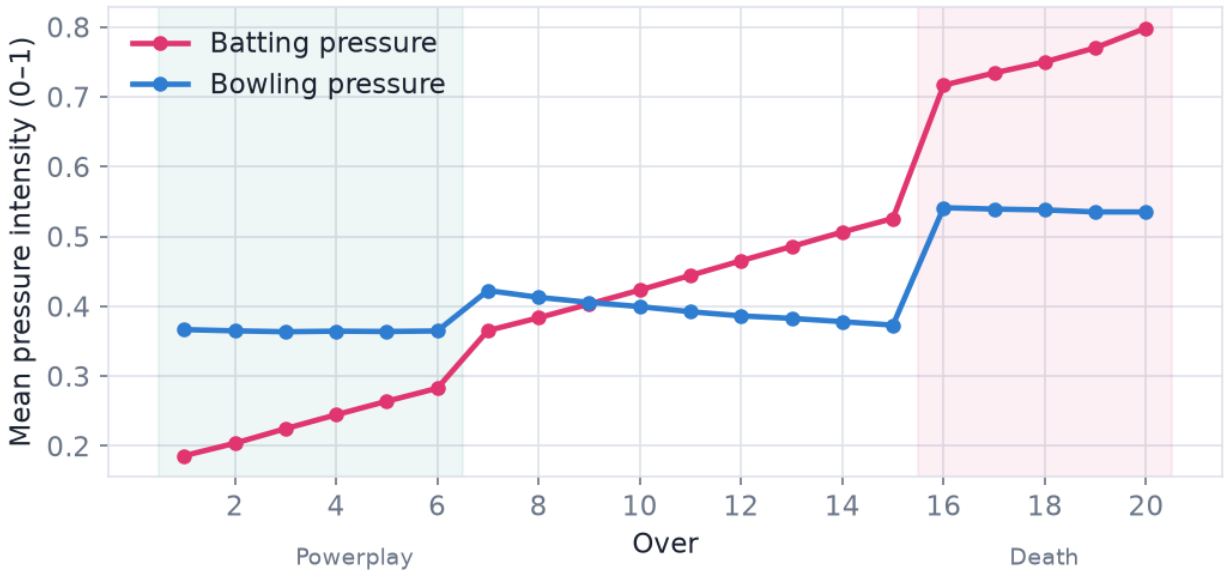
4 · Data Cleaning & the Extras Puzzle

The single most important data-quality finding: **extras1_runs is not a number of runs — it is a category code**, one-to-one with extras_type (6 = legal ball, 2 = wide, 3 = leg-bye, 4 = bye, 1 = no-ball, 5 = other, 7 = penalty). Summing it would be meaningless. We confirmed that **runs** is **runs off the bat** (every six has runs = 6, every four runs = 4, with zero contradictions; all wides/byes/leg-byes carry runs = 0). Therefore:

- **Batting metrics are exact** — strike rate, boundary %, dot %, dismissals all come straight from reliable fields.
- **Bowler economy is reliable** — byes and leg-byes are not charged to the bowler, so a bowler's runs = off-bat + wides + no-balls (+ recorded extra runs).
- **Team totals / required run rate are approximate** — bye/leg-bye magnitudes beyond 1 are not stored. Validated against 356 result margins: median error 0, 266 within ±3 runs; the large misses are all DLS rain games, which we exclude from chase logic.
- **Bowler wickets** exclude run-outs, retired-out and obstruction (609 of 10,881); 10,272 are credited. **Legal balls** exclude wides, no-balls and the re-bowled ‘other’ extra. All aggregation keys on player_id, since some names map to two IDs.

5 · Pressure Definition

Pressure is a weighted blend of normalised signals, each in [0,1], combined differently for batting and bowling. The chart below shows the mean intensity by over: batting pressure climbs steadily and jumps at the death; bowling pressure is flatter but peaks when batters attack — exactly the asymmetry we want.



Batting pressure rises with

- **Innings phase** — death overs carry the highest stake, powerplay the lowest.
- **Wickets already lost** — a rebuild after a collapse is higher pressure.
- **Required run rate** (chases) — a demanding chase is high pressure, but a *hopeless* chase (needing >3 runs/ball with few resources) is dampened, since garbage-time hitting is not a fair clutch test.

Bowling pressure rises with

- **Death overs**, where margins are smallest.
- **Batting momentum** — a high current run rate with wickets in hand means the bowler must stem the flow.
- **Defending a gettable chase** — when the batting side is cruising (low required rate), the bowler is under most pressure to break through.

A delivery is classified ‘pressure’ when its intensity ≥ 0.55 (configurable; swept in §12). This yields 33% batting-pressure and 19% bowling-pressure deliveries.

6 · Match-State Reconstruction

For every delivery we reconstruct the state before the ball, chronologically: score so far, wickets fallen, wickets in hand, legal balls bowled and remaining, and — in the second innings — the target (first-innings total + 1), runs required, required run rate and current run rate. Balls remaining are clamped to ≥ 0; DLS matches are flagged and excluded from chase-based pressure. Correctness is covered by unit tests on hand-built innings (score, wickets, legal-ball counting, RRR, and a no-future-leakage invariant).

7 · Batter Performance Metrics

On each delivery set (pressure and the player's own baseline) we compute: balls faced, runs, strike rate, average (runs per dismissal, counting only the batter's own dismissals), dot-ball %, boundary %, and match counts. The headline comparison is pressure vs baseline strike rate and survival.

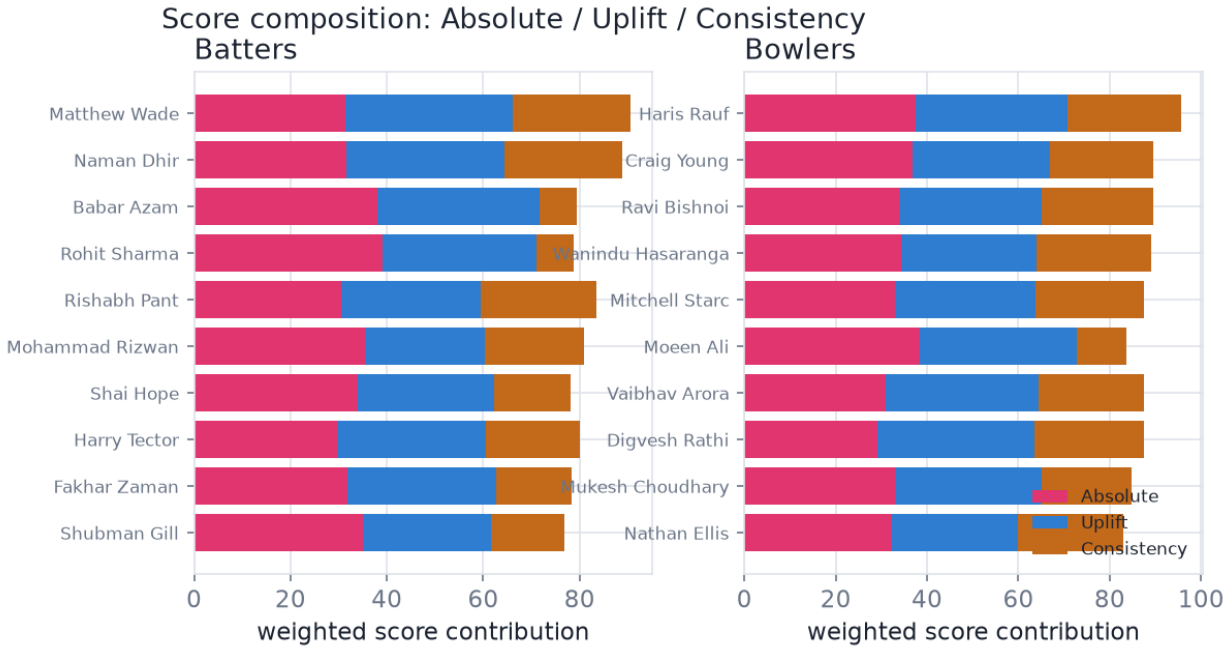
8 · Bowler Performance Metrics

Per delivery set: legal balls, runs conceded (bowler-charged), credited wickets, economy, strike rate, wicket rate per over, dot-ball % and boundary % conceded — each compared against the bowler's own baseline.

9 · Player Rating Methodology

Score = 0.50 · Absolute + 0.30 · Uplift + 0.20 · Consistency (each a 0–1 percentile within the eligible pool, ×100). Absolute carries the most weight so a genuinely high-level pressure performance is rewarded on its own merit, while uplift and consistency keep the measure honest about who *lifts* when it matters:

- **Absolute (50%)** — performance under pressure, *format-adjusted* against the league's pressure par for the formats the player actually played (so IPL and T20I are compared fairly). Batters: pressure SR + survival; bowlers: economy-vs-par + wicket rate + dot %.
- **Uplift (30%)** — pressure minus the player's **context-matched** baseline (own non-pressure level in the same innings phases, shrunk toward their overall baseline), then **empirical-Bayes shrunk** toward zero by pressure sample size. Context-matching removes the death-over illusion from the raw uplift.
- **Consistency (20%)** — the share of a player's pressure *matches* in which they delivered. Batters: SR within 10% of their (context-matched) baseline, a tolerance so high-baseline players are not penalised for a strong-but-slightly-below-average innings. Bowlers: a **graded** blend of economy improvement (45%), wicket contribution (30%) and dot-ball rate (25%), replacing the old ‘economy ≤ baseline OR one wicket’ rule that a single wicket could fully satisfy.



10 · Sample Size, Eligibility & Reliability

Eligibility (configurable): batters need ≥ 75 pressure balls, ≥ 5 pressure matches, ≥ 150 baseline balls; bowlers ≥ 60 pressure legal balls, ≥ 5 pressure matches, ≥ 120 baseline balls. Shrinkage priors (60 balls for batters, 48 for bowlers) pull small-sample uplifts toward zero. We never tune thresholds to force a particular player in.

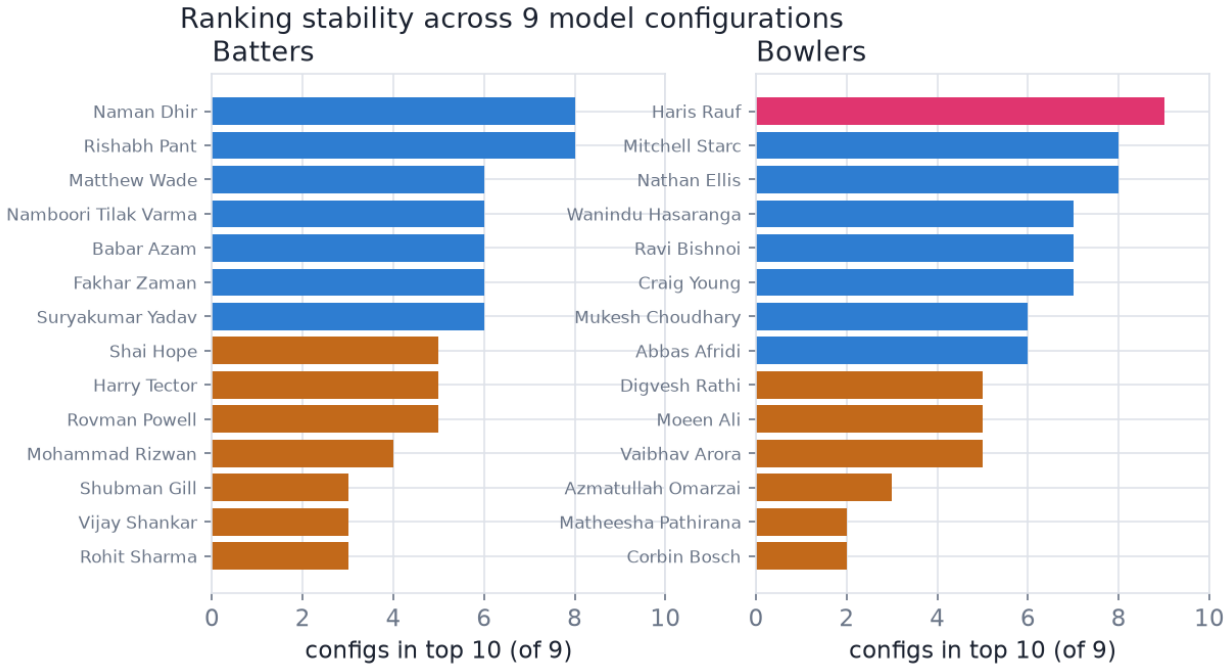
11 · IPL and T20I Comparison

Both formats are combined, but baselines are computed per player (their own non-pressure balls in the formats they played) and the absolute component is adjusted against format-specific pressure par (IPL pressure SR ≈ 153 vs T20I ≈ 138; IPL pressure economy ≈ 10.6 vs T20I ≈ 10.0). Two sensitivity configs — IPL-weighting and a strict-sample cut — check that associate-nation inflation does not drive the picks.

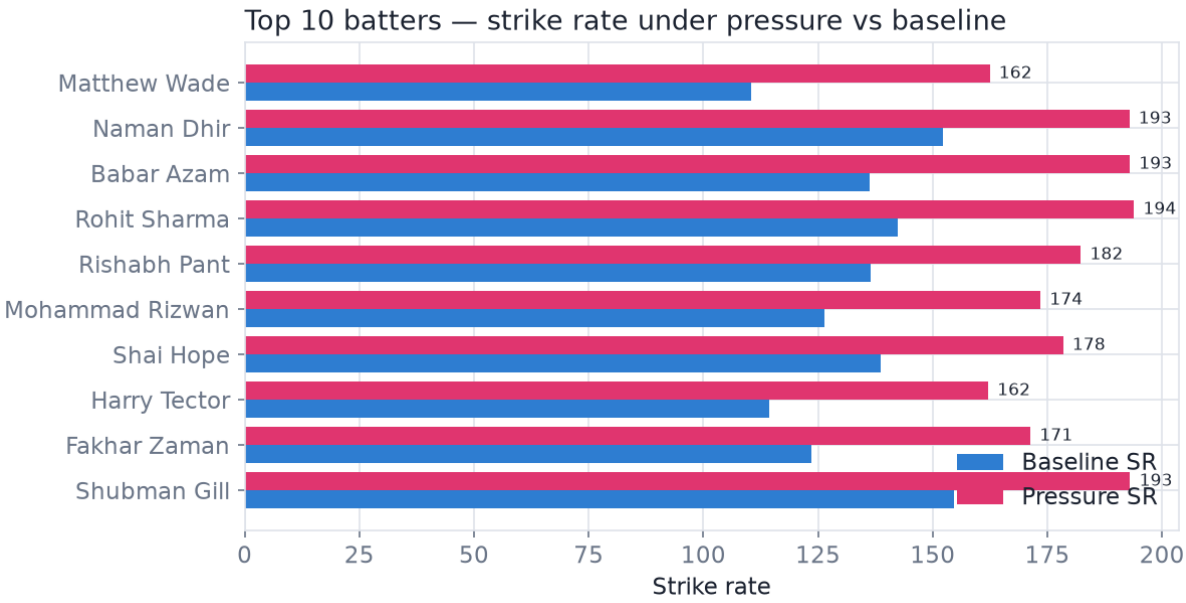
12 · Sensitivity & Ranking Stability

We re-ran the entire rating under nine reasonable configurations: three pressure thresholds (0.50/0.55/0.60), two alternative batting definitions (chase-centric; death-and-collapse with no RRR) and two bowling definitions

(death-heavy; defend-heavy), a strict-sample variant, and three weight mixes (uplift-, absolute- and consistency-heavy). The bars show how often each player stayed in the top 10.



13 · Top 10 Batters — Evidence



#	Batter	Score	Rel.	Pr. balls	Pr. mts	Pressure SR	Base SR	SR upl.	Pr. avg
1	**Matthew Wade**	88.6	67%	229	19	162	110	+52	53
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7	**Shai Hope**	79.3	56%	116	12	178	139	+40	34
8	**Harry Tector**	79.2	56%	182	16	162	114	+48	30
9	**Fakhar Zaman**	78.8	67%	129	12	171	124	+48	32
10	**Shubman Gill**	78.8	33%	202	17	193	155	+38	39

Why each batter is here (all figures from the supplied data)

1. Matthew Wade — pressure strike rate **162** vs baseline 110 (**+52**), averaging 53 under pressure over 229 pressure balls in 19 matches; delivered in 94% of qualifying pressure matches; stays top-10 in 67% of configurations.

2. Naman Dhir — pressure strike rate **193** vs baseline 152 (**+41**), averaging 32 under pressure over 267 pressure balls in 25 matches; delivered in 90% of qualifying pressure matches; stays top-10 in 89% of configurations.

3. Babar Azam — pressure strike rate **193** vs baseline 136 (**+57**), averaging 44 under pressure over 114 pressure balls in 8 matches; delivered in 75% of qualifying pressure matches; stays top-10 in 67% of configurations.

4. Rohit Sharma — pressure strike rate **194** vs baseline 142 (**+52**), averaging 80 under pressure over 82 pressure balls in 5 matches; delivered in 100% of qualifying pressure matches; stays top-10 in 33% of configurations.

5. Rishabh Pant — pressure strike rate **182** vs baseline 137 (**+46**), averaging 33 under pressure over 339 pressure balls in 31 matches; delivered in 86% of qualifying pressure matches; stays top-10 in 89% of configurations.

6. Mohammad Rizwan — pressure strike rate **174** vs baseline 126 (**+47**), averaging 41 under pressure over 117 pressure balls in 9 matches; delivered in 100% of qualifying pressure matches; stays top-10 in 44% of configurations.

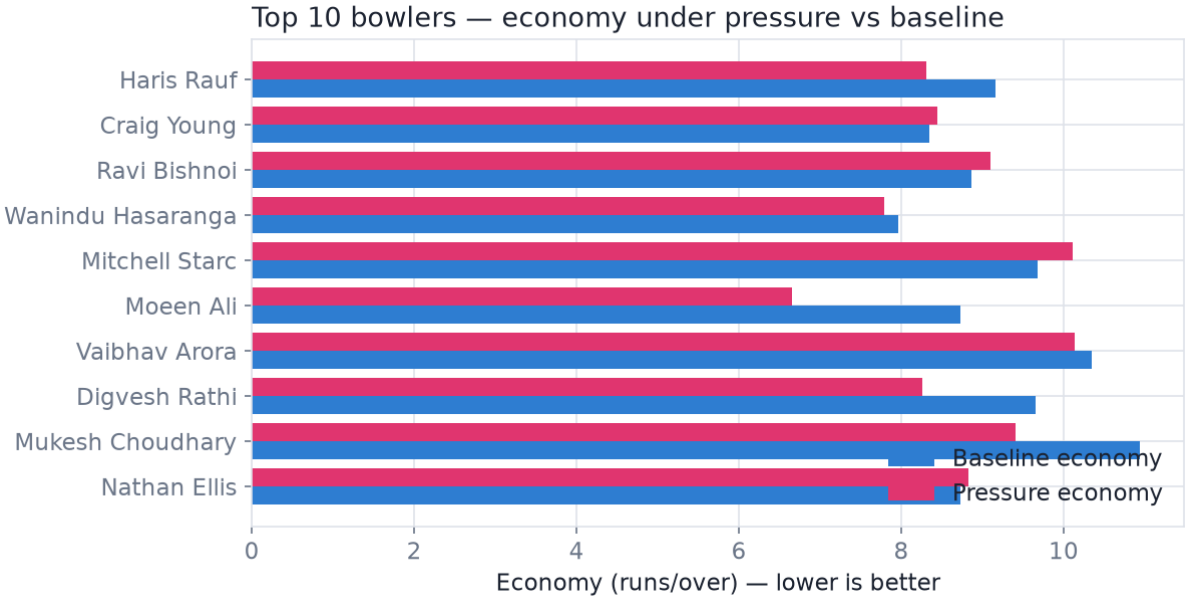
7. Shai Hope — pressure strike rate **178** vs baseline 139 (**+40**), averaging 34 under pressure over 116 pressure balls in 12 matches; delivered in 88% of qualifying pressure matches; stays top-10 in 56% of configurations.

8. Harry Tector — pressure strike rate **162** vs baseline 114 (**+48**), averaging 30 under pressure over 182 pressure balls in 16 matches; delivered in 83% of qualifying pressure matches; stays top-10 in 56% of configurations.

9. Fakhar Zaman — pressure strike rate **171** vs baseline 124 (**+48**), averaging 32 under pressure over 129 pressure balls in 12 matches; delivered in 88% of qualifying pressure matches; stays top-10 in 67% of configurations.

10. Shubman Gill — pressure strike rate **193** vs baseline 155 (**+38**), averaging 39 under pressure over 202 pressure balls in 17 matches; delivered in 73% of qualifying pressure matches; stays top-10 in 33% of configurations.

14 · Top 10 Bowlers — Evidence



#	Bowler	Type	Score	Rel.	Pr. balls	Pr. mts	Pr. wkts	Pressure econ	Base econ	Wkts/ over
1	**Haris Rauf**	Pace	95.3	100%	283	29	28	8.31	9.16	0.59
2	**Craig Young**	Pace	89.9	78%	98	11	9	8.45	8.34	0.55
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5	**Mitchell Starc**	Pace	86.6	89%	171	20	19	10.11	9.68	0.67
6	**Moeen Ali**	Spin	86.1	56%	73	11	8	6.66	8.73	0.66
7	**Vaibhav Arora**	Pace	85.9	56%	171	22	15	10.14	10.34	0.53
8	**Digvesh Rathi**	Spin	85.1	56%	146	14	9	8.26	9.65	0.37
9	**Mukesh Choudhary**	Pace	84.5	67%	97	13	7	9.40	10.94	0.43
10	**Nathan Ellis**	Pace	82.4	89%	193	21	13	8.83	8.72	0.40

Why each bowler is here (all figures from the supplied data)

1. Haris Rauf (pace) — economy **8.31** vs baseline 9.16, 0.59 credited wickets/over and 42% dot balls under pressure, over 283 pressure legal balls in 29 matches; stays top-10 in 100% of configurations.

2. Craig Young (pace) — economy 8.45 (vs 8.34 baseline), 0.55 credited wickets/over and 41% dot balls under pressure, over 98 pressure legal balls in 11 matches; stays top-10 in 78% of configurations.

3. Ravi Bishnoi (spin) — economy 9.09 (vs 8.86 baseline), 0.48 credited wickets/over and 39% dot balls under pressure, over 223 pressure legal balls in 28 matches; stays top-10 in 78% of configurations.

4. Wanindu Hasaranga (spin) — economy **7.79** vs baseline 7.96, 0.46 credited wickets/over and 34% dot balls under pressure, over 432 pressure legal balls in 37 matches; stays top-10 in 78% of configurations.

5. Mitchell Starc (pace) — economy 10.11 (vs 9.68 baseline), 0.67 credited wickets/over and 40% dot balls under pressure, over 171 pressure legal balls in 20 matches; stays top-10 in 89% of configurations.

6. Moeen Ali (spin) — economy **6.66** vs baseline 8.73, 0.66 credited wickets/over and 37% dot balls under pressure, over 73 pressure legal balls in 11 matches; stays top-10 in 56% of configurations.

7. Vaibhav Arora (pace) — economy **10.14** vs baseline 10.34, 0.53 credited wickets/over and 36% dot balls under pressure, over 171 pressure legal balls in 22 matches; stays top-10 in 56% of configurations.

8. Digvesh Rathi (spin) — economy **8.26** vs baseline 9.65, 0.37 credited wickets/over and 32% dot balls under pressure, over 146 pressure legal balls in 14 matches; stays top-10 in 56% of configurations.

9. Mukesh Choudhary (pace) — economy **9.40** vs baseline 10.94, 0.43 credited wickets/over and 41% dot balls under pressure, over 97 pressure legal balls in 13 matches; stays top-10 in 67% of configurations.

10. Nathan Ellis (pace) — economy 8.83 (vs 8.72 baseline), 0.40 credited wickets/over and 40% dot balls under pressure, over 193 pressure legal balls in 21 matches; stays top-10 in 89% of configurations.

15 · IPL-Availability — an Optional, Secondary View

The problem statement does not ask for a nationality filter, so the **primary rankings in this report apply none** — they are the pure statistical result. Separately, because Rajasthan Royals is an IPL franchise, we provide an *optional* view that removes players an IPL side cannot sign. Nationality is **inferred** from the national team a player represents in T20Is — a data-only heuristic that is **not independently verified** (players with no T20I record cannot be classified), which is why it is presented only as a secondary lens. The scoring model is identical; this view drops the flagged players and re-ranks.

Removed from the IPL-availability view — batters: Babar Azam, Mohammad Rizwan, Fakhar Zaman, Usman Khan, Salman Agha, Iftikhar Ahmed, Hasan Nawaz, Mohammad Haris.

Removed from the IPL-availability view — bowlers: Haris Rauf, Abbas Afridi, Sufiyan Muqeem, Abrar Ahmed, Naseem Shah, Shadab Khan, Shaheen Afridi, Mohammad Nawaz, Mohammad Wasim.

Full IPL-available top-10s are in outputs/batter_rankings_ipl_available.csv and the dashboard.

16 · Model Validation

- **Unit-tested cricket logic** — hand-built innings verify score/wicket reconstruction, legal-ball counting, bowler wicket credit, required run rate, and a no-future-leakage invariant (11/11 pass).
- **Innings totals cross-checked** against 356 result margins (median error 0; 266 within ±3 runs; large misses are DLS games).
- **Pressure definition is meaningful** — mean intensity rises monotonically through the innings and differs by role, matching cricketing intuition.
- **Small samples are shrunk, not rewarded** — the sample-size-vs-score plots show no runaway effect from low-ball players.
- **Rankings are stable** — the strongest picks appear in the top 10 across nearly all nine configurations.

17 · Limitations

- **Approximate targets** — bye/leg-bye magnitudes are not stored, so required run rate is accurate to ±1–3 runs; DLS matches are excluded from chase logic. Coarse pressure bands make this immaterial to classification.
- **Pressure is a modelling choice**, not an official RR definition; we mitigate with nine configurations and per-player reliability.
- **T20I strength varies** by opponent; baselines are format-adjusted but some associate-nation inflation can remain — checked via IPL-weighted and strict-sample configs.
- **No fitted win-probability model** — contest closeness is proxied by required run rate, wickets and phase.
- **Availability filter** uses nationality inferred from T20I appearances; a handful of uncapped players have no T20I record and are treated as IPL-eligible.

18 · Conclusion

The framework turns the vague idea of ‘clutch’ into a reproducible, delivery-level measurement: a transparent pressure index, self-referenced performance, a balanced and shrinkage-protected rating, and a robustness check across nine model variants. The resulting top-10 batters and bowlers are not reputation picks — every name is backed by its own pressure-versus-baseline numbers and a reliability score, and every figure in this report is traceable to the supplied data via the accompanying code (run_pipeline.py) and interactive dashboard.